

PAYMENTS SERVICE — DESIGN SPEC

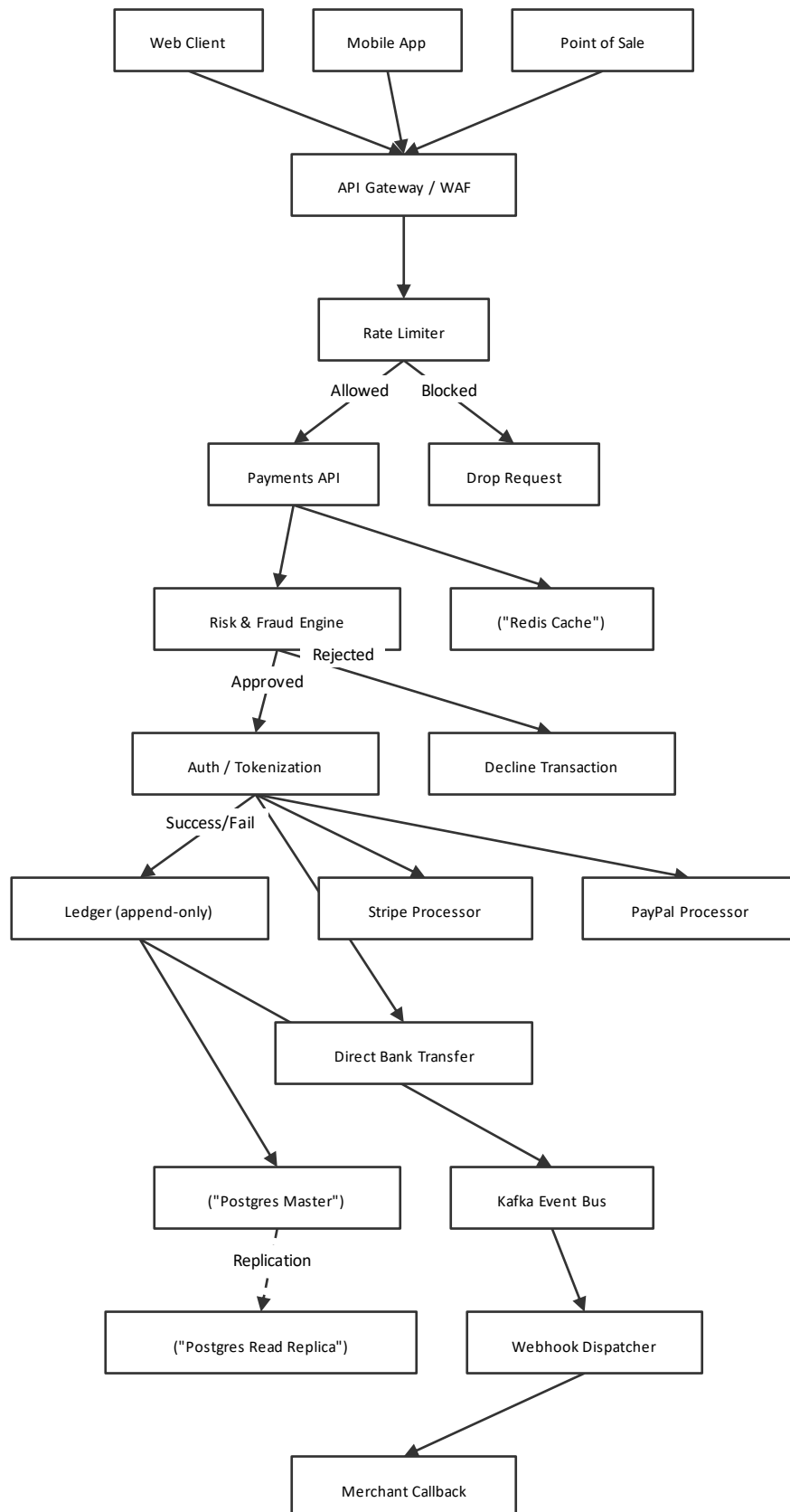
small, auditable service that authorises card payments and records every state transition.

A

NOTE

This document is illustrative. Amounts and identifiers are examples, not production values.

Architecture



Request flow

1. The client posts a charge with an idempotency key.
2. The API tokenizes the card and calls the processor.
3. Every transition is appended to the **ledger** — nothing is ever mutated in place.
4. A webhook notifies the merchant asynchronously.

```
{  
  "idempotency key": "chg_9f2a...",  
  "amount": 4200,  
  "currency": "AUD",  
  "capture": true  
}
```

State machine

State	Next	Trigger
created	authorized	processor approves
authorized	captured	capture requested
authorized	voided	timeout / cancel
captured	refunded	refund issued

WARNING

Idempotency keys expire after 24h. Re-using an expired key starts a **new** charge.